Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Real-Time Video Chat with Vision-Language Models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Technology Field 2 is optional*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW		
Concept of Proposed Technology	We are developing technology that enables real-time video chat and question answering (Q&A) over online video streams by building and optimizing large vision-language model (VLM) systems. Naive deployments of VLMs for live video processing incur substantial latency due to the computational cost of large models, while memory overhead further limits the ability to handle long-duration video streams. Our project addresses these limitations through careful system-level and model-level optimizations. The system ingests video streams in real time, supports low-latency natural language interaction, and efficiently processes long video streams spanning hours. Users can pose queries such as "What color is the traffic light?" or "Summarize the last 20 seconds," and receive accurate responses within interactive latency bounds. This technology has the potential for broad impact across multiple industries. Smart wearable devices (e.g., Meta Ray-Ban AI glasses) can provide JARVIS-like visual assistance with significantly reduced latency. Autonomous robots and self-driving vehicles can make faster and more informed decisions through increased video processing throughput. Smart manufacturing and factory			

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Rationale for Joint R&D Project	automation can benefit from real-time anomaly detection and situational awareness. Building production-ready and highly optimized vision-language model systems is a non-trivial undertaking that requires deep expertise across artificial intelligence, computer vision, and computer systems. Many Korean SMEs face practical constraints in investing the time, financial resources, and engineering talent required for such advanced R&D efforts. The NSAI (Networked Systems and AI) Lab at UIUC brings extensive experience in designing and optimizing video AI systems for real-time applications. By partnering with us, Korean SMEs can accelerate the deployment of cutting-edge research into their products, reducing R&D risk and time to market. At the same time, this collaboration enables our lab to validate and refine these systems under real-world operational constraints and domain-specific requirements that are difficult to replicate in academic environments.
Keywords	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Vision-Language Models 2. Real-Time Communications 3. Video Analytics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	Although the project is still at an early stage, our research group already possesses a set of working research prototypes and core system components for real-time VLM inference. We are actively analyzing state-of-the-art VLM systems [3,4,5] and have developed prototypes that support VLM-based online inference using industry-grade model serving frameworks (e.g., vLLM). Building on this foundation, we are investigating novel system architectures and optimization techniques to improve inference latency, throughput, and memory efficiency. Our objective is to advance these prototypes into robust, production-ready systems capable of supporting real-time question answering over long-duration video streams.			
R&D Target Specifications	The objective is to retain the high Q&A accuracy of current state-of-the-art frameworks while substantially reducing response latency, as measured on widely used benchmarks such as OVO-Bench [1] and StreamingBench [2].			
R&D Plan	1. Analysis and Requirement Scoping a. Analyze state-of-the-art vision-language model (VLM) systems and identify key performance bottlenecks. b. Establish evaluation metrics and develop benchmarking pipelines. c. Scope system requirements and identify target use cases, deployment scenarios, and operational constraints. 2. Design & Prototyping a. Design VLM system architectures optimized for streaming inference by leveraging video stream characteristics and minimizing redundant computation. b. Prototype and iterate on multiple candidate designs to select the most effective architecture. 3. System Optimization and Robustness a. Optimize system performance under realistic deployment constraints, including heterogeneous compute environments (e.g., GPUs), variable video quality, network conditions, and query rates. b. Evaluate scalability across multiple concurrent video streams. 4. Validation a. Integrate the developed system into commercial platforms. b. Conduct user-driven evaluations to assess usability, responsiveness, and overall system effectiveness.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	NSAI at UIUC will serve as the R&D lead for the project, providing technical guidance and domain expertise to ensure smooth integration of the developed technology into the SME partner's commercialization strategy. NSAI will review project requirements and objectives, adapt the system architecture to meet these needs, and evaluate deployments to ensure technical soundness and real-world viability. Throughout the project, NSAI will contribute research personnel, experimental infrastructure, and ongoing technical guidance.
Expected R&R of Korean SME Partner	The Korean SME is expected to provide clear requirements, domain expertise, and real-world datasets relevant to its products or services. The SME will be responsible for supporting integration of the system into existing pipelines and for providing continuous feedback on system performance and usability. In addition, on the commercialization side, the SME will define product vision, manage operational considerations, and address legal and regulatory requirements.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Through this joint R&D project, Korean SMEs will gain the ability to apply advanced AI capabilities to real-time question answering or video chat over online video streams. The proposed technology will substantially reduce inference latency and memory overhead while maintaining high response accuracy, enabling practical deployment in real-time operational settings. From a commercialization perspective, the technology can be directly integrated into a wide range of products, including wearable devices, smart cameras, autonomous systems, and factory automation software. This collaboration will accelerate time to market and help Korean SMEs remain globally competitive with large corporations that possess dedicated and specialized R&D teams. In the long term, we hope this project will strengthen industry-academic collaboration between Korea and the United States. Korea is already a global leader in manufacturing and has significant potential to further enhance productivity through automation and AI. However, a persistent gap remains in translating AI research into reliable, production-ready systems. We believe this project will help bridge that gap by enabling the practical productionization of AI technologies, contributing to sustainable industrial innovation and economic growth in Korea.

6. References

[1] Li, Y., Niu, J., Miao, Z., Ge, C., Zhou, Y., He, Q., Dong, X., Duan, H., Ding, S., Qian, R., & Zhang, P. (2025). *OVO-Bench: How Far is Your Video-LLMs from Real-World Online Video Understanding?* arXiv:2501.05510.

[2] Lin, J., Fang, Z., Chen, C., Wan, Z., Luo, F., Li, P., Liu, Y., & Sun, M. (2024). *StreamingBench: Assessing the Gap for MLLMs to Achieve Streaming Video Understanding.* arXiv:2411.03628.

[3] Di, S., Yu, Z., Zhang, G., Li, H., Zhong, T., Cheng, H., Li, B., He, W., Shu, F., & Jiang, H. (2025). *Streaming video question-answering with in-context video KV-cache retrieval*. arXiv:2503.00540.

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[5] Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., Ge, W., Guo, Z., Huang, Q., Huang, J., Huang, F., Hui, B., Jiang, S., Li, Z., Li, M., Li, K., Lin, Z., Lin, J., Liu, X., Liu, J., Liu, C., Liu, Y., Liu, D., Liu, S., Lu, D., Luo, R., Lv, C., Men, R., Meng, L., Ren, X., Ren, X., Song, S., Sun, Y., Tang, J., Tu, J., Wan, J., Wang, P., Wang, P., Wang, Q., Wang, Y., Xie, T., Xu, Y., Xu, H., Xu, J., Yang, Z., Yang, M., Yang, J., Yang, A., Yu, B., Zhang, F., Zhang, H., Zhang, X., Zheng, B., Zhong, H., Zhou, J., Zhou, F., Zhou, J., Zhu, Y., & Zhu, K. (2025). *Qwen3-VL technical report*. arXiv:2511.21631.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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